Production Closure of Derivative-Free Causal Integration in VigilODE: Exact Constants, Full-Space Certificates, Adversarial Profiles, and Local Qualification

VigilODE research and implementation audit

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Starting from PR91, we independently pursue structural stage verification, scalar homotopy implementation, and selective adapter decisions. No newer PR was found in the initial remote survey. An exact represented-stage identity replaces repeated mean-field full-space verification by an outward enclosure of cost $O(sn + s^2)$, while retaining every state component and charging failed parallel work. A 2,048-variable, one-worker trajectory has a paired median speedup of 2.063 and 62.52% fewer guest instructions than PR91. Scalar homotopy removes allocation overhead without changing numerical bits; a proved rational inverse predictor reduces cubic polynomial evaluations from 1,659 to 1,024 under the same physical certificate. The two routes reduce guest instructions by 36.26% and 39.01%, respectively. The adapter's 10.95% work reduction on one stiff case does not establish a wall-time win; its reentrant nonlinear alternative remains slower than RHS-only in every paired block. Exact integer comparison and failure accounting are therefore kept separate from future-regime optimality. Three, one, and three correction passes were used for the respective research tasks. A strict serialized production facade, independent error oracles, native regression checks, source-bound execution receipts, and a 16-node local development DAG provide a reviewable deployment setup. General nonlinear closure/order, temporal residence guarantees, and actual deployment-host release qualification remain explicit obligations.

I. SCOPE, PRIOR PRS, AND RESEARCH DISCIPLINE

The implementation baseline is commit `206298b4068d923bb47c7160b111b7439098660c`, the head of draft PR91 when the work began. The initial GitHub survey inspected the recent 30 PRs and found no PR newer than 91. PR90 and PR91 remained open drafts, so this

change is prepared on a separate branch stacked above PR91; it does not merge either predecessor or alter the default branch. The exact survey receipt and source identities accompany the repository.

The predecessor research is actively reused. PR65's counted common-W corrections warn that sequential forward solves can double work without shortening the critical path. PR74 and PR75 supply the coupled raw-U target, ordered parallel dispatch, original-residual screening, and charged speculation. PR77 and the derivative-free originals supply physical collective/transverse closures. PR90 provides careful prefix-work evidence and the distinction between residual and physical error. PR91 provides the exact two-channel mean-field reduction, the dyadic cubic physical-error plan, exact-input causal reuse, and strict entry selection. Existing SPD04 least-squares workspace reuse is already enabled in the facade; a known nonnormal Givens differential failure remains an explicit constraint on any replacement. These contributions are controls and premises, not new claims of the present report.

Three parallel agents completed separate theory and implementation tracks, with parent integration and complementary independent native-agent reviews. The PhysMath Research Loop version 1.0.0 organizes R1–R10 and C1–C10. Every task records its original target, assumptions, alternatives, exact derivation, numerical counterprobes, source-bound implementation, actual tests, and bounded corrections. Shared native builds, timings, and profiles are serialized by a single lock. Reviewers share project context; their reviews are not blinded or heterogeneous-model replications.

The research search used SciSpace for discovery, Exa and primary papers for source checking, and Wolfram for symbolic identities and independent inequality checks. No retrieved theorem was treated as a blanket convergence guarantee for this solver. In particular, modern derivative-free Anderson acceleration for monotone nonlinear equations has its own projection and monotonicity assumptions [2]. Historical parallel iterated RK work separates nonstiff fixed-point iteration from stiff parallel constructions [3, 4]. Neither an implicit corrector's stability nor a paper's favorable timings transfer automatically to an arbitrarily truncated causal or homotopy loop.

TABLE I. Separate research tasks and correction budgets. The initial attempt is recorded separately from a corrective pass. Packaging and environment repairs are labeled, rather than silently counted as numerical tuning.

Task	Diagnosed target and implemented response	Corrections
Structural	Failed-work loss; repeated full-space residual arithmetic; adverse scalar cost. Charged API, proved enclosure, exact $n = 1$ degeneration.	3/3
Homotopy	Vec/station overhead; costly cubic radicals. Scalar representation and algebraically derived rational predictor with original certification.	1/3
Adapter	Discarded useful pilot proposals; empirical return margin; expensive residence in nonlinear regimes. Exact comparisons and causal proposal reuse, with negative outcomes retained.	3/3

II. EVIDENCE HIERARCHY AND MAIN MEASURED OUTCOMES

The report distinguishes four different objects: a real-arithmetic identity; a floating-point outward enclosure under declared arithmetic assumptions; an independent experiment against an original physical target; and a timing measurement on one host. A smaller projected residual is insufficient for a full-state conclusion. Fewer callback units need not imply fewer machine instructions or lower elapsed time. A stage residual bound alone is not a trajectory accuracy certificate.

The strongest structural timing observation is the one-worker $n = 2048$ enclosure: $2.063\times$, range $[1.226, 3.614]$, under the same 64-step trajectory and physical cap 10^{-6}. The all-grid states match PR91 bit for bit. The two-worker range crosses one, so it supplies no general parallel speed guarantee. A direct fused implementation reduces instructions but has a negative final one-worker timing median; it remains a measured alternative rather than the recommended universal choice.

For the nonlinear cubic cohort, generic/scalar and generic/rational medians are 1.474 and 1.670, with all five observed ratios above one. The scalar/rational ranges cross one in

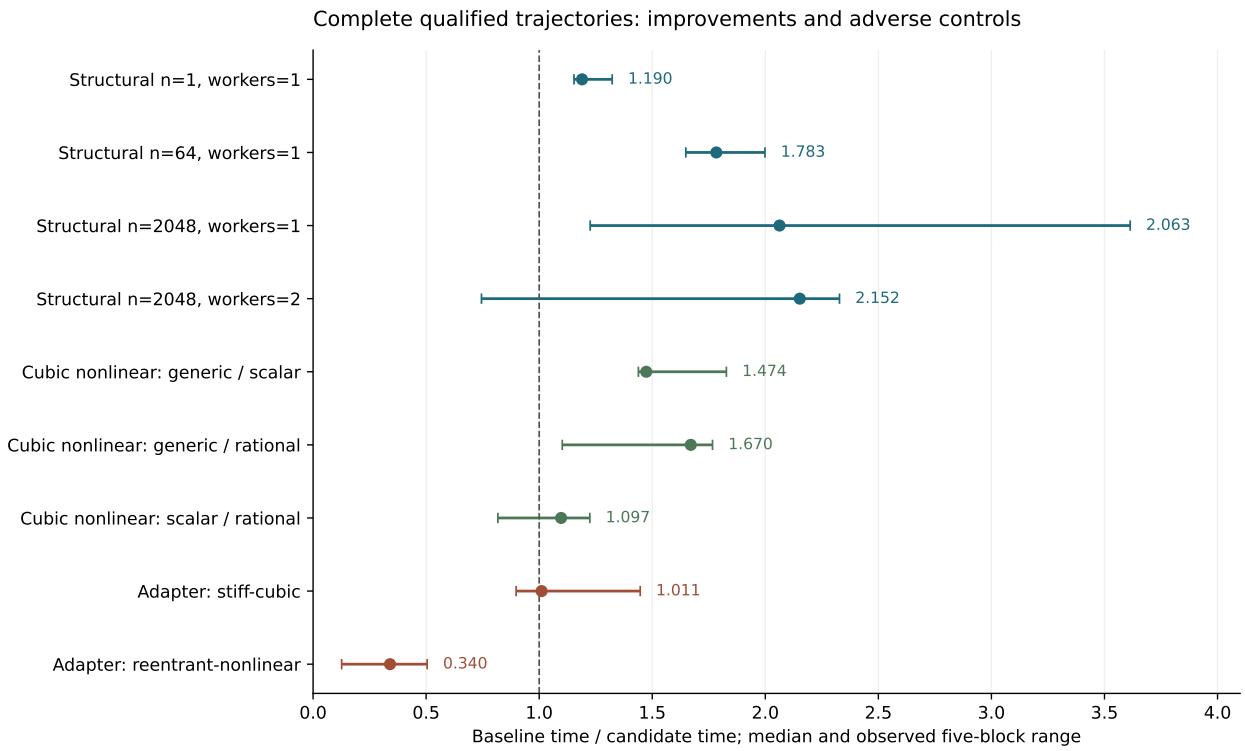


FIG. 1. Measured complete-trajectory speed ratios. Dots are medians of five ABBA/BAAB blocks; bars are the observed minimum and maximum, not confidence intervals. Different rows have their explicitly named baselines. The adapter stiff row compares PR91 with observed proposal reuse; its reentrant row compares RHS-only with observed proposal reuse and sentinel.

both cubic cohorts. We therefore expose both strong candidates without installing a fitted runtime threshold. The recommended scalar Picard representation is a conservative opt-in implementation choice; the legacy public entry still selects the generic homotopy. Rational prediction is a separately selectable model-specific alternative. The original outward physical certificate is retained.

The adverse adapter profile is actionable: the adapter dispatcher itself has negligible self instruction cost, but its decision to remain implicit incurs norm, QR, orthogonalization and GMRES costs. Optimizing a few dispatcher branches cannot remove this residence loss. Future work must separately improve the mode arithmetic and derive when historical evidence is admissible for later regimes. The current public deployment default is therefore persistent RHS-only; selective supplied JVP is opt-in.

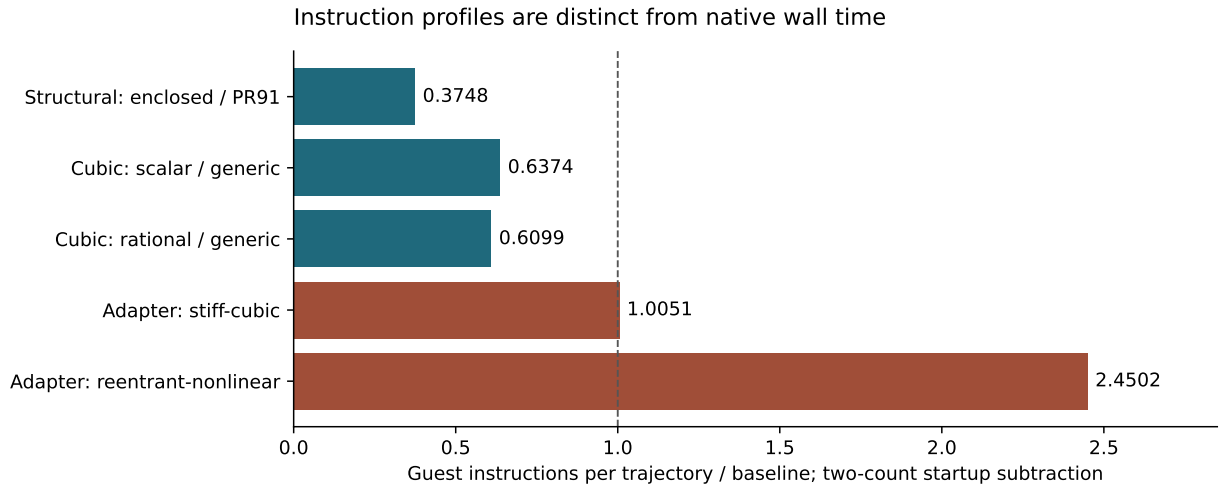


FIG. 2. Startup-subtracted Callgrind instruction ratios from the same executable within each comparison. These guest instruction counts are neither hardware cycles nor estimates of end-to-end native wall time.

III. STRUCTURAL MEAN-FIELD CLOSURE: CHARGED FAILURES AND FULL-SPACE RESIDUAL ENCLOSURES

For the immutable affine model

$$f(y) = b - dy + \kappa\,\mathrm{mean}(y)\mathbf{1}, \qquad d \geq \kappa \geq 0, \tag{1}$$

PR91 reduced the original eight-stage raw-U target to two scalar transfer recurrences. The present change preserves every represented stage and endpoint in the successful comparison fixtures. It closes the failed-attempt work contract and replaces repeated full-state verification with a rounding-aware factorization. The model remains derivative-free: no state Jacobian, JVP, state matrix factorization, or Newton update is requested. The shared tableau loader retains its process-lifetime fixed 8×8 coefficient inversion; this metadata setup is outside step counters and warmed before paired timing.

A. Original target and exact residual decomposition

Let $P = \mathbf{1}\mathbf{1}^T/n$, $Q = I - P$, $J = -dQ - (d - \kappa)P$, and $a = h\gamma$, with $h > 0$ and the fixed tableau's $\gamma > 0$. The full original residual is

$$\mathcal{R}_i(U) = (I - aJ)U_i - af(y) - \sum_{j<i}(\gamma c_{ij} I + a a_{ij} J)U_j. \tag{2}$$

All represented input and tableau coefficients are interpreted as exact reals; the enclosure includes the real product $h\gamma$ and real difference $d - \kappa$.

a. Theorem S1 (represented-stage decomposition). For arbitrary x, μ and transfers t_i, v_i, write

$$U_i = t_i(x - \mu\mathbf{1}) + v_i\mu\mathbf{1} + \Delta_i, \quad \varepsilon_f = x - f(y), \quad m = \mathrm{mean}(x). \tag{3}$$

Define

$$\rho_i(r; z) = (1 + ar)z_i - \sum_{j<i}(\gamma c_{ij} - ara_{ij})z_j - a, \tag{4}$$

$$\rho_{ti} = \rho_i(d; t), \quad \rho_{ci} = \rho_i(d - \kappa; v), \quad \ell_i = t_i + \sum_{j<i} a_{ij}t_j. \tag{5}$$

Then

$$\begin{aligned}\mathcal{R}_i(U) = {} & \rho_{ti}(x - \mu\mathbf{1}) + \rho_{ci}\mu\mathbf{1} + a\kappa\ell_i(\mu - m)\mathbf{1} + a\varepsilon_f \\ & + (I - aJ)\Delta_i - \sum_{j<i}(\gamma c_{ij}I + aa_{ij}J)\Delta_j.\end{aligned} \tag{6}$$

Substitution and $Jx = -dx + \kappa m\mathbf{1}$ prove the identity. In particular, the transverse factor is $x - \mu\mathbf{1}$ with the actually computed μ; its discrepancy from the exact projector is explicitly retained. A Wolfram symbolic substitution independently returned a zero vector. The proof applies for arbitrary dimension and arbitrary full-dimensional Δ_i.

b. Theorem S2 (full original-space bound). The exact induced infinity norm of $tQ + cP$ is

$$\nu_n(t, c) = \left|t + \frac{c - t}{n}\right| + (n - 1)\left|\frac{c - t}{n}\right|. \tag{7}$$

Set

$$w_{ii} = \nu_n(1 + ad, 1 + a(d - \kappa)), \tag{8}$$

$$w_{ij} = \nu_n(ada_{ij} - \gamma c_{ij}, a(d - \kappa)a_{ij} - \gamma c_{ij}), \quad j < i. \tag{9}$$

If $\|\Delta_j\|_\infty \le \delta_j$, $\|\varepsilon_f\|_\infty \le \epsilon$, $|\mu - m| \le \zeta$, and $\|x - \mu\mathbf{1}\|_\infty \le X$, then

$$\|\mathcal{R}_i(U)\|_\infty \le |\rho_{ti}|X + |\rho_{ci}|\,|\mu| + |a\kappa\ell_i|\zeta + a\epsilon + \sum_{j\le i} w_{ij}\delta_j. \tag{10}$$

This follows directly from S1 and the induced norm inequality. Every scalar coefficient, block norm and nonnegative final accumulation is enclosed outward. The bound covers all $8n$ original residual components at cost $O(sn + s^2)$ instead of $O(s^2n)$. Only the individual block norm is asserted sharp; the combined rounding envelope is generally conservative.

B. Rounding constants and scalar specialization

a. Theorem S3 (operation-wise rounding enclosure). The arithmetic assumptions are binary64 round-to-nearest, gradual underflow, finite intermediates, and no unsafe reassociation. Let $r(z)$ be half the binade spacing at a finite rounded result, rounded upward to the least positive subnormal when necessary. This is the round-to-nearest error radius; it is derived from the exponent, with no fitted factor.

For the existing scaled Neumaier mean, put $s = \mathrm{fl}(1/n)$, $v_i = \mathrm{fl}(sx_i)$ and let C_i be the accumulated compensation. FastTwoSum supplies exact local addition residuals [1]. Upward sums

$$A \geq \sum_i |v_i|, \quad E_p \geq \sum_i r(v_i), \quad E_c \geq \sum_i r(C_i), \quad \theta \geq r(s)/s \tag{11}$$

yield

$$|\widehat{m} - \mathrm{mean}(x)| \leq E_p + \theta(A + E_p) + E_c + r(\widehat{m}). \tag{12}$$

Telescoping the exact local corrections proves the accumulation part; the remaining two terms bound scaled-product and reciprocal error. The enclosure plan requires an exactly represented positive integer dimension at most 2^{53}.

For an actual stage lift $p = \mathrm{fl}(t_i\,\mathrm{fl}(x_k - \mu))$, $q = \mathrm{fl}(v_i\mu)$ and $U_{ik} = \mathrm{fl}(p+q)$, the sum of the four local error bounds gives

$$\|\Delta_i\|_\infty \leq \delta_i := r(\max_k |U_{ik}|) + r(\max_k |p_k|) + r(q) + |t_i| r(\max_k |\,\mathrm{fl}(x_k - \mu)|). \tag{13}$$

The affine RHS is enclosed from its represented product, subtraction, shared mean term and final addition in the same way.

b. Theorem S4 (exact scalar degeneration). For $n = 1$, $P = I$ and $\mu = x$ exactly. Consequently the real represented stage is one rounded product $U_i = \mathrm{fl}(v_i x)$, together with exact signed-zero operations retained for bit compatibility. With $u = 2^{-53}$ and $\eta = 2^{-1074}$,

$$|\,\mathrm{fl}(z) - z| \leq u|z| + \eta/2. \tag{14}$$

The unrepresentable positive $\eta/2$ is enclosed upward by η. Therefore

$$|\mathcal{R}_i(U)| \leq \left(|\rho_{ci}| + u \sum_{j \leq i} w_{ij}|v_j|\right)|x| + \eta \sum_{j \leq i} w_{ij} + a\epsilon. \tag{15}$$

The two sums are plan-only quantities and are precomputed outward, reducing per-step scalar verification to $O(s)$. The special case is the exact projector degeneration, not a measured dimension cutoff.

TABLE II. Final paired full-trajectory timings versus PR91. Values are medians of five ABBA/BAAB blocks with their observed ranges, not confidence intervals. Model allocation, plan construction, stages, screening, endpoint and return allocations are timed.

n	workers	fused direct	factorized enclosure
1	1	1.011 [0.737, 1.231]	1.190 [1.154, 1.323]
64	1	1.168 [0.915, 1.266]	1.783 [1.649, 1.998]
2048	1	0.840 [0.619, 1.220]	2.063 [1.226, 3.614]
2048	2	1.213 [1.032, 1.276]	2.152 [0.745, 2.328]

C. Failure accounting and comparison design

The new charged methods return `StructuralFailure{error,work}`. Scheduler closures return each stage result as data, all jobs are joined, their actual work is accumulated, and the first failure in stage order is returned. A failed job cannot cancel another scheduled job's accounting. Pre-attempt refusals retain zero work. Legacy method signatures remain available; adaptive clients use the charged variants.

Two implementations were compared against the byte-identical PR91 source in the same executable. The fused direct screen removes two full-state temporary vectors while preserving the original arithmetic order. The factorized screen supplies an outward bound and refuses when that bound exceeds the original requested limit. Full 64-step trajectories retain the original physical cap 10^{-6}, same-target cap 10^{-11} and residual cap 10^{-9}. Every observed state was bit-identical to PR91.

At $n = 2048$, complete-trajectory RHS evaluations fell from 576 to 64, physical linear actions from 512 to zero, and causal full-state accumulations from 7,340,032 to zero. All 1,048,576 stage residual components remain covered; 262,144 mean components and 1,048,576 lift components receive explicit rounding enclosures. The maximum grid error is unchanged at 2.398×10^{-10}.

The initially unfavorable scalar result was retained: before the last correction, the enclosure gave 0.392× and the charged fused implementation 0.694× in five paired blocks. The exact $P = I$ specialization and algebraic aggregation of batch work address that diagnosed cost. The final scalar enclosure gives 1.190×. This is a post-diagnostic improvement; final

large-case fused timing remains uncertain and its negative median is reported.

Eleven distinct native contracts pass (five reused structural contracts and six new production contracts). Independent exact-rational evaluation of the full original equations confirms 168 stage bounds across 21 fixtures and 728 residual components, including large cancellation, nonrepresentable rate differences and subnormals. The 8,184 exponent-wide checks of round-to-nearest radii are reused after the scalar update. Three correction passes were used: crate-local control packaging, a portable integer dimension guard, and the diagnosed scalar/accounting optimization. No further solver tuning was performed.

A final Callgrind 3.22.0 profile used two and six complete trajectories per arm at $n = 2048$, one worker, and subtracted startup via $(I_r(6) - I_r(2))/4$. All six runs exited zero with unchanged source and binary. Instructions per trajectory were 347,516,244.75 for PR91, 309,511,284.25 for fused direct, and 130,256,982 for the enclosure: reductions of 10.94% and 62.52%, respectively. These guest instruction counts are reported separately from native wall time; the fused timing median remains negative in the final large single-worker sample. Protocol, commands, raw profiles and hashes are retained in `instruction_profile/`.

The certificate concerns original stage residuals. A complete outward physical trajectory certificate still requires outward propagation constants, temporal truncation and endpoint-rounding contributions. These are explicit local follow-up obligations; the current measured trajectory accuracy and stage certificate are separate evidence.

IV. RHS-ONLY SCALAR CLOSURE AND CERTIFIED INVERSE CANDIDATES

The sealed consumer solves $y' = -ky - gy^3$, with $k > 0$, $g \geq 0$ and $y(0) = Y \geq 0$. Its PR91 dyadic plan, finite-step truncation coefficient $C = (k + 3gY^2)^2(kY + gY^3)/8$, and outward physical certificate are unchanged. Candidate generation is now selectable: the original generic homotopy, a stack-valued scalar specialization, a positive-sum cubic inverse, and a rational inverse predictor. The generic public entry point is retained.

a. Exact inverse representation. For the midpoint equation $am + bm^3 = y$, put

$$a = 1 + hk/2, \qquad b = hg/2, \qquad \beta = by^2/a^3, \tag{16}$$

$$z = \tfrac{3}{2}\sqrt{3\beta}, \qquad u = (z + \sqrt{1 + z^2})^{1/3}, \qquad s = \frac{3}{u^2 + 1 + u^{-2}}. \tag{17}$$

Then $m = (y/a)s$ is the unique real solution. Indeed, the polynomial is strictly increasing. For $\beta > 0$, $u^3 - u^{-3} = 2z$ and

$$(u - u^{-1})(u^2 + 1 + u^{-2}) = 2z \tag{18}$$

give $s = (u - u^{-1})/\sqrt{3\beta}$. The identity $(u - u^{-1})^3 + 3(u - u^{-1}) = u^3 - u^{-3}$ proves $s + \beta s^3 = 1$. At $\beta = 0$, the positive-sum formula directly gives $s = 1$. The existing contraction domain gives $0 \le \beta < 1/3$ and $0 \le z < 3/2$. No subtraction of nearly equal cube roots is required.

b. Rational alternative and its defect. The inverse series starts $1 - \beta + 3\beta^2$. Matching a [1/1] rational expression uniquely determines

$$r(\beta) = \frac{1 + 2\beta}{1 + 3\beta}, \qquad r + \beta r^3 - 1 = \frac{\beta^3(3 + 8\beta)}{(1 + 3\beta)^3}. \tag{19}$$

The right-hand side is nonnegative for every $\beta \ge 0$. Since the residual derivative is $1+3\beta s^2 \ge 1$, its unique positive root s_* satisfies

$$0 \le r - s_* \le \frac{\beta^3(3 + 8\beta)}{(1 + 3\beta)^3}. \tag{20}$$

These coefficients follow from algebra; none was fitted to timing data. Wolfram symbolic evaluation and 151 exact rational cells for each identity provide independent checks of the displayed formulas.

c. Original-target certificate. Both inverse candidates form the increment as $\widehat{K} = -h\phi(\widehat{m})$, avoiding the subtraction $2(\widehat{m} - y)$. Their floating-point accuracy is not assumed. Acceptance evaluates the original $F(K) = K + h\phi(y + K/2)$ in outward intervals. For every real K, $F'(K) \ge D = 1 + hk/2$, hence

$$|\widehat{K} - K_*| \le |F(\widehat{K})|/D. \tag{21}$$

Adding the enclosed represented-update displacement preserves the previous nonexpansive physical-error induction. A missed budget invokes charged scalar Picard and, if needed, the existing monotone bisection. A rounding-floor refusal preserves the accepted prefix. Predictor counts are reported separately from ordinary and interval polynomial evaluations.

d. Representation and measured costs. The scalar specialization preserves operation order, seed, actual residual screen, and all charged callback evaluations. Forty combinations of model, mode and seed policy preserve every returned state and error-bound bit. Twenty-three distinct native contracts pass. The final 24 accuracy arms, containing 5,404 returned

observations, satisfy the unchanged native outward bound under an independent Decimal80 exact-ODE oracle.

For $k = 1$, $g = 1/2$, $Y = T = 1$, Table III compares warm Picard against the rational predictor. Scalar Picard has the same work and output bits as generic Picard. The final outward bound in the tight rational case is $4.46606\,10^{-6}$. Same-binary Callgrind differences between 20 and 40 complete trajectories give instruction slopes 3,182,149.3 (generic), 2,028,310 (scalar), 2,061,615 (algebraic), and 1,940,807 (rational). Scalar and rational reduce this measure by 36.26% and 39.01%, respectively. Rational is 4.31% below the strong scalar implementation. These are guest instructions with process startup subtracted, not hardware cycles.

TABLE III. Identical physical accuracy plans. Each rational step also performs one separately counted rational inverse calculation.

ϵ	N	Picard RHS+interval	Rational RHS+interval	Rational max. error
10^{-3}	64	198	128	$1.43850\,10^{-5}$
10^{-4}	128	462	256	$3.59356\,10^{-6}$
10^{-5}	512	1659	1024	$2.24475\,10^{-7}$

Five paired ABBA/BAAB wall-time blocks include full output allocation and all interval verification. Generic/scalar median ratios are 1.728 for the baseline cubic and 1.474 for the nonlinear-dominated fixture. Corresponding generic/rational medians are 1.514 and 1.670. Scalar/rational medians are 0.970 and 1.097, with observed ranges crossing unity; local hardware selection between these strong alternatives remains warranted. The expensive cubic inverse was the diagnosed adverse alternative. Exactly one corrective loop introduced the rational predictor, and all initial negative measurements remain.

The original physical certificate is retained in every comparison. Its `Interval::mul` instruction count is unchanged and accounts for 55.06% of the rational profile. Improving the finite-step truncation estimate and vector-valued homotopy workspace are separate subsequent research tasks.

V. ADAPTER SELECTION AND MODE ARITHMETIC ARE SEPARATE COSTS

The adapter now offers exact integer return selection and causal transport of an observed entry-step proposal. Both new flags default to false. The general production configuration retains persistent RHS-only execution; supplied-JVP fallback is an explicitly qualified option. The measured adverse cases below are retained in the release evidence.

Let

$$W = N_f + N_{Jv} + N_{f_t} + nN_J + N_{\text{solve}}. \tag{22}$$

This is the existing accounting objective. For a completed explicit candidate of work E and a same-window implicit incumbent of work I, strict improvement is exactly $E < I$. If the implicit replay is incomplete, its charged nonfatal prefix L proves the same preference when $E < L \leq I$. The implementation checks the complete candidate and finite window identity, then compares checked `u128` sums; ties retain the incumbent.

a. Exact stopping constants. The native callback guard charges the crossing callback before stopping at $L_{\text{cb}} >$ cap. Consequently, a completed explicit candidate requires an implicit-comparator cap E: crossing proves $E < L_{\text{cb}} \leq I$. A productive implicit sentinel of cost I instead bounds its explicit challenger by $I - 1$. Crossing then makes a strict explicit win impossible. These integer boundaries follow from the comparison objective rather than an empirical gain margin. Since a native DOPRI attempt performs at most seven RHS calls, the sentinel has

$$W_{\text{sentinel}} \leq 2I + 6, \qquad I \geq 1. \tag{23}$$

This is an overhead bound relative to the current implicit window, whose cost can itself be unfavorable.

b. Causal proposal transport. An earlier nonfatal pilot with $m > 0$ accepted steps from a to $c > a$ provides $\bar{h} = (c-a)/m$. A later pilot may use $\max(h_0, h_{\text{checkpoint}}, \bar{h})$, clipped to its native replay interval, while retaining the later accepted physical state and every native error and work test. This transports a past scalar proposal; route selection still requires a fresh successful cost comparison. Zero-accepted and fatal pilots provide no proposal. Forty-four actual pilot initializations satisfy this causal formula in the recorded trajectories.

TABLE IV. Whole-trajectory charged work. All discarded pilots and return probes are included. "Observed" transports the accepted pilot-step mean.

Case	RHS-only	PR91	Observed	Observed+sentinel
Easy cubic	79	79	79	79
Stiff cubic	2347	2958	2634	2634
Relaxing nonlinear	10375	11407	16331	12195
Reentrant nonlinear	17113	18600	27546	24440

TABLE V. Paired timing and instruction slopes for complete qualified trajectories. Ranges are five observed block ranges, not confidence intervals.

Comparison	Instruction change	Median speedup [range]
Stiff: PR91 to observed	+0.505%	1.0106 [0.8981,1.4469]
Reentrant: RHS-only to sentinel	+145.02%	0.3404 [0.1265,0.5046]

c. Nonlinear exact-state tests. For the autonomous augmentation $c' = 1$ and

$$y_j' = -\lambda(c)(e_j + e_j^3/2) + g'(c), \qquad e_j = y_j - g(c), \tag{24}$$

separation with $z = e_j^{-2}$ gives $z' = 2\lambda z + \lambda$ and hence

$$e_j(t) = \frac{e_j(0)e^{-\Lambda(t)}}{\sqrt{1 + e_j(0)^2[1 - e^{-2\Lambda(t)}]/2}}, \qquad \Lambda(t) = \int_0^t \lambda(s)\, ds. \tag{25}$$

The constant, relaxing and reentrant rates have explicit Λ through $\log\cosh$. Independent Decimal90 evaluation checks all 17,440 returned state components. Exact rational Laurent-polynomial arithmetic verifies the inverse-square transformation and the supplied directional Jacobian action. All 36 arms pass their original accuracy and callback-accounting contracts.

The original 20-arm test produced no selective switches: a completed pilot could cost 204 against 192 explicit units, and its useful accepted step was discarded. The observed-step correction enables entry, but Table IV shows that this can increase total work after a regime change. Sentinel returns recover part of that loss. Exact and ratio-based return decisions coincide on this physical cohort; integer near-tie tests distinguish their decision boundaries separately.

Five ABBA/BAAB blocks, recorded before the final branch-style cleanup, use one fixed measured executable and 64 full solves per process for stiff cubic and 16 for reentrant relaxation. Model and result allocation are timed; exact-state analysis and JSON serialization follow the timer. Callgrind differences between four and twelve repeated solves remove fixed startup/output cost. The stiff-cubic work decrease of 10.95% does not produce a stable measured time improvement. The reentrant selective route is slower in every paired block. In its twelve-solve profile, dispatch function `drive_windows` has only 0.0225% self instructions, whereas `safe_l2`, pivoted QR, two-pass MGS and GMRES account for 9.77%, 3.72%, 3.53% and 2.63%, respectively. These self costs identify mode arithmetic separately from the adapter's choice of which mode to execute.

Thirty-five distinct native contracts pass. Three correction passes were used: observed-step transport after the negative initial cohort, then initialization of its new field in an existing private test fixture, and finally two equivalent let-chain rewrites required by production CI. The final rewrite preserves left-to-right short-circuit evaluation and changes no policy, numerical constant or fixture. Historical timing and profiling retain their original source and executable identities. Fatal-callback priority, accepted-prefix preservation, strict integer boundaries, forbidden derivative work, configuration serialization and all prior relevant return contracts remain tested. The final 36-arm correctness audit exactly reproduces all 17,440 state components, counters and segment decisions in the earlier records; only elapsed time is excluded from the comparison and the original missing default-false option is normalized. Existing high-precision oracle evidence is reused on these identical inputs. No final-source paired timing or instruction measurement is inferred from that equality. Further development must address cost intervals for small Krylov arithmetic, quantified residence loss under changing regimes, and complete native-session transfer after winning explicit probes.

VI. WHAT IS A MATHEMATICAL GAP, AND WHAT IS CODE OPTIMIZATION?

The following division is based on the claims required by each route. It does not infer a strong impossibility theorem from a finite failed experiment. A restricted construction can remain useful while a broader claim is unsupported.

A. Structural closure and a physical trajectory theorem

The affine mean-field quotient is exact because the declared operator preserves the span of the collective and transverse projectors. Its arbitrary full-state forcing is retained. Repeated full-state verification was consequently an implementation bottleneck with a constructive exact factorization. Rounding, however, means that a computed mean is not the exact projector and a lifted stage is not the exact two-channel state. Theorem S1 supplies the missing terms, and S2–S4 turn them into an executable enclosure. Discarding these terms would have been a mathematical defect in a certificate, not merely an inefficient implementation.

The new stage certificate still does not establish an outward physical trajectory bound. The old accumulated algebraic budget is evaluated in ordinary binary64, and time truncation, endpoint addition and observation time errors have independent contributions. A variable-step extension must derive the adjoint/product residual transport, bound every rounding term, and state its operator-commutation conditions. A nonlinear extension must bound the closure remainder on an explicit invariant domain. PR-N03 carries these proof obligations before an adaptive consumer can claim a physical certificate.

B. Causal parallelism and high-order nonlinear carriers

The finite triangular target admits exact forward substitution and finite causal propagation. These statements establish neither cheap parallel execution nor the high-order conditions of a changing nonlinear carrier. The inherited general causal path still uses a frozen JVP basis and small projected linear algebra; calling it universally derivative- or matrix-free would be incorrect. Exact cache identity and buffer reuse are implementation tasks. Proving original-space residual amplification, closure remainders, and order under carrier rebuilding is a theory task.

The relevant performance comparison includes total work and critical path. Independent stage or model jobs can execute concurrently, but communication, validation, pool startup and the best forward solve are part of the comparison. PR-N05 keeps that strong serial baseline and an explicit work/span contract. The existing successful structural special case is evidence for this restricted algebra, not a general proof that sequential RK dependencies can always be removed profitably.

C. Root finding, certificates, and remaining accuracy cost

The scalar representation is a direct code optimization: fixed arithmetic order and 40 mode/seed/model comparisons establish bit parity. The rational inverse is a constructive algebraic alternative whose coefficients and defect are proved. Its acceptance continues to use the original equation, so no unproved libm cube-root accuracy is trusted. The specific monotone cubic model supplies existence, uniqueness, a bracket and a residual-to-root constant. A generic nonlinear homotopy needs its own branch, domain, monotonicity or inverse-control assumptions. Without them, the finite-budget failure contract remains explicit.

After root-cost reduction, interval multiplication accounts for 55.06% of the rational instruction profile. Removing certificate work without a replacement proof would relax the scientific target. PR-N02 instead seeks a tighter finite-step defect along the actual decay envelope, an outward sum of propagated error, and the smallest admitted grid. This targets the number of certified steps, while generic vector workspace reuse in PR-N04 targets allocation. Their savings must be attributed separately.

D. Selection evidence and future regimes

Exact integer comparison proves a current same-target work preference. The $2I + 6$ sentinel theorem bounds current-window overhead relative to its implicit incumbent. Neither statement bounds the incumbent's future cost. The measured reentrant loss demonstrates the gap between the selector's proxy and actual full-trajectory cost in this cohort. It is not a theorem that derivative-free or selective methods are universally inferior.

A defensible future-regime claim must assume a quantitative variation, dwell-time or model-domain condition, define evidence epochs, and charge exploration and invalidated probes. PR-N06 is distinct from the arithmetic cost model in PR-N01. It seeks a finite-horizon residence/exploration bound and budget-derived revalidation rules. An empirical multiplier without such a theorem would not meet the requested exact-constant standard.

E. Constants and caller choices

New method constants have an explicit origin: 2 and 3 in the rational inverse arise from series matching; $1+hk/2$ is the exact strong-monotonicity lower bound; the structural block weight is an induced norm; 2^{-53} and 2^{-1074} are binary64 arithmetic constants; the adapter's $I-1$, E and $2I+6$ follow from integer strictness and native attempt overshoot. None is a fitted hyperparameter.

This does not erase legitimate user inputs. Physical tolerances, initial step proposals, thread count, attempt/resource caps and observation windows belong to the client or execution environment. Existing controller and pressure/backoff policy constants are preserved operational choices, not newly proved universal optima. The JSON templates use one declared smoke fixture and label these inputs accordingly. A claim that every inherited hyperparameter has been eliminated would exceed the work performed.

VII. PRODUCTION CONFIGURATION AND FAILURE-PRESERVING EXECUTION

The additive public API exports `ProductionConfig`, `ProductionSolver`, and schema version one. The default factory selects persistent RHS-only execution. The selective factory accepts only a supplied JVP, requires replay-safe callbacks, preserves the explicit session, and installs strict entry and return evidence. It does not automatically enable observed-step reuse or one-window sentinel probing. Unsupported derivative permissions and incomplete configuration are rejected before callbacks run.

Strict deserialization covers the top level, switching configuration, nested options and absolute tolerance. Unknown fields such as a misspelled step limit are rejected. The legacy general `Options` deserialization behavior is unchanged; production uses a strict intermediate representation. Native feature validation continues to govern mass matrices, nonautonomous time derivatives, dense output, events and tolerance combinations. This is an endpoint-oriented smooth forward identity-mass setup, not a promise that every experimental route supports every facade feature.

The result retains success, reached time, method segments, actual callback work, accounting completeness and an accepted prefix on numerical failure. Callers inspect these

fields instead of equating an outer successful return with a completed integration. Charged structural APIs are required for new adaptive or retrying consumers; legacy wrappers intentionally discard only the failure work payload for signature compatibility. A callback panic under an aborting process profile cannot fulfill a returned partial-result contract. Expected model failures should use the declared callback error.

The executable `production_from_json` emits complete templates and reads them back. On $n = 16$, $y' = -ry$, $[0, 1]$, four combinations of RHS-only/selective and $r = 1, 1000$ pass the original 10^{-6} all-returned-state cap and atomic work accounting. The RHS-only cases make zero JVP calls. The stiff selective case records 1,117 RHS and 426 JVP calls against 2,377 RHS calls in the default case; its observed accuracy is 1.0654×10^{-9}. This controlled smoke result does not override the adverse nonlinear adapter evidence.

VIII. VALIDATION, REPRODUCIBILITY, AND INTERPRETATION LIMITS

The full facade and coupled native test sets pass 297 and 127 tests, respectively, with zero failed or ignored tests in the final invocation after the integration-accounting repair described below. The narrower lane counts (11 structural, 23 homotopy, 35 adapter and five production-setup contracts) overlap those complete package sets and are not added again. All-target compilation, warning-free rustdoc and two doctests are checked separately. Final integration gate receipts and any later format-only qualification are authoritative in `evidence/VALIDATION_SUMMARY.json`; they distinguish tests from compilation, linting and publishing.

The broader five-package invocation exposed an inherited lagged-GMRES accounting defect after 1,312 passing tests, one failure and 45 ignored tests. The solver-level counter already charged attempted operator applications, whereas the physical work hook charged successful completions. With A entered applications and P charged physical applications, the missing physical work is exactly $A - P$ for representable counters. Reconciliation against the attempted-operator count had incorrectly yielded zero for a failed JVP. The local repair separates these ledgers, leaving callbacks, numerical arithmetic, tolerances and the original failing assertion unchanged. The original regression and successful dense/JVP contracts then pass. The recorded post-repair scope covers eight reachable lagged targets and the complete facade/coupled sets: 464 distinct tests pass, none fail, and four measurement tests

remain ignored. This is not a fresh successful five-package sweep. Unrelated exact-oracle evidence is reused through source and route identity. Full deployment release qualification remains a local gate.

Independent oracles include 728 exact full structural residual components and 168 stage bounds across 21 fixtures; 5,404 cubic grid observations against Decimal80; and 17,440 nonlinear adapter state components against Decimal90. Exact-rational symbolic checks and Wolfram receipts accompany the proofs. The unchanged structural rounding-radius routine reuses 8,184 exponent/mantissa checks. Finite checks supplement the displayed proofs; they do not establish unrestricted floating-point correctness by sampling.

The shared host uses an Intel Xeon Platinum 8573C, nine logical CPUs in the affinity set, an eight-core CPU quota and an 8-GiB memory limit. Rust 1.94.1 and the locked dependency graph are pinned. Cloud measurements use opt-level three, no LTO and 16 codegen units. Deployment qualification uses the existing release profile with fat LTO and one codegen unit. Those profiles and their binary hashes must not be mixed. Native timings are descriptive paired observations on a shared host. Oracle evaluation and JSON serialization are outside the recorded timer; the relevant model, plan, solve, validation and return allocation costs are included as stated in each protocol. Thread-pool construction is reported separately.

The new command recorder binds actual argv, environment, source/input hashes, executable identity, stdout/stderr hashes, timeout and exit status. Source discovery is repeated after execution, so new files also invalidate a qualification. A successful command with changed source is recorded as stale and returns failure. Owned process groups are cleaned on timeout, cancellation and direct-child completion before the suite lock is released. Seven actual toy-process contracts exercise success, nonzero exit, missing executable, modified/new input and resistant-grandchild timeout/cancellation. Code review covers the spawn and lock races; those tests do not exhaust all scheduler interleavings.

The initial negative cohorts, compile-time fixture failure, profiler environment preflight failure, and unfavorable final measurements are retained. Environment repairs do not become numerical improvement claims. Raw evidence is stored with a per-member hash manifest and reproducible archive, alongside the final source manifest. The package verifier checks identity and DAG consistency; its pass is a packaging result, not a mathematical proof or a deployment certificate.

TABLE VI. Future local PR contracts. Every research PR starts with its own unchanged target and at most three corrective passes. The JSON plan includes code targets, expected artifacts, reuse links and explicit acceptance tests.

Node	Prerequisite	Purpose and required evidence
PR-N01	L-HARDWARE	Small Krylov norm/QR/MGS cost; retain existing LS workspace; prove true-residual and rank/breakdown conditions.
PR-N02	L-RELEASE	Finite-step cubic bound on the decay envelope; outward error sum and smallest admitted grid.
PR-N03	L-RELEASE	Variable-step transport, temporal truncation and endpoint rounding; charged adaptive consumption.
PR-N04	L-RELEASE	Complete session/epoch transfer and generic vector homotopy workspace with preserved arithmetic.
PR-N05	L-HARDWARE, N04	Nonlinear closure/order and total work/span; comparison against forward substitution.
PR-N06	L-HARDWARE, N01	Temporal residence and exploration loss under declared variation or dwell assumptions.

IX. LOCAL DEVELOPMENT DAG AND PR CONTRACTS

The machine-readable plan contains 16 nodes and eight executable suites. The completed remote survey feeds the three research lanes; those feed production integration, cloud validation, and publication. Local release verification precedes hardware measurements. Four development branches can then pursue independent scoped work. Native timing and profiling stay serialized even when proofs and disjoint edits run concurrently.

L-RELEASE pins the published commit and records actual release/LTO tests, docs, four smokes, compiler, libm, CPU and source identities. Ignored tests retain a separate

status. L-HARDWARE then measures paired complete trajectories and instruction profiles while preserving scalar and nonlinear negative controls. R-QUALIFY admits only actual client/model/provider combinations whose original accuracy and resource contracts pass these two gates.

The runner is dry-run by default. A user-specified fresh output directory receives a suite receipt and per-command source snapshots, stdout/stderr and exit records. Its POSIX/WSL lock serializes builds and measurements. The local benchmark suite covers structural $n = 1, 64, 2048$, bounded worker counts, paired cubic alternatives and the adverse adapter. The profile suite uses two repetition counts to subtract startup. The handoff pins an isolated worktree to the published commit; source and evidence manifests verify that identity.

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